Distracted Driver Detection using Deep Learning

Group Number: 26

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1. Revised Problem Statement

Distracted driving causes over 3,000 fatalities annually in the US. This project develops a real-time distracted driver detection system using deep learning to classify 10 driver behaviors (safe driving, texting, phone usage, drinking, reaching, makeup, talking to passengers) from dashboard camera images. We compare three CNN architectures (EfficientNet-B0, ResNet18, MobileNetV3) optimized for edge deployment, with novel contributions in zero-shot domain generalization analysis and quantitative CAM quality evaluation using the Pointing Game metric.

2. Project Adjustments & Team Progress

Added Novel Contribution 1: Zero-shot domain generalization analysis comparing State Farm and AUC datasets. Added Novel Contribution 2: CAM quality evaluation using Pointing Game metric. Project Completion: ~85%

3. Progress Since Proposal

Architecture: Implemented subject-aware train/val split to prevent data leakage. Used ImageNet pretrained EfficientNet-B0 with fine-tuning. Applied augmentation: RandomResizedCrop, ColorJitter, RandomErasing. Training: 5 epochs, batch 32, AdamW, lr=1e-4.

Performance Metrics:

Model	Accuracy	Macro F1	Params	Latency
EfficientNet-B0	88.06%	86.85%	4.02M	13.9ms

Per-Class Accuracy:

Safe	75.3%	Text R	90.9%	Phone R	98.7%	Text L	98.3%	Phone L	98.1%
Radio	95.0%	Drink	92.7%	Reach	90.1%	Makeup	85.6%	Passenger	43.9%

4. Model Improvements & Deployment

Identified "Talking to Passenger" as hardest class (43.9%) - confused with Safe Driving. Planned fix: focal loss and class-weighted sampling. Deployment: ONNX export complete, 72 FPS throughput, Grad-CAM visualization ready.

5. Challenges & Solutions

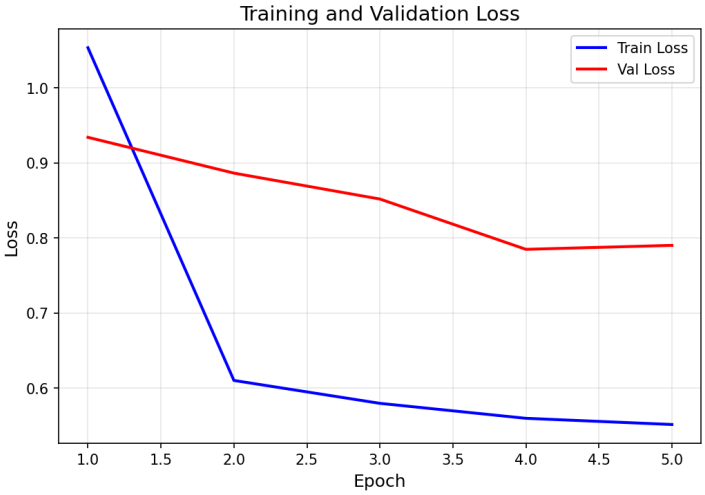
- Data Leakage Risk: Random split could place same driver in both sets. Solution: Subject-aware splitting using driver IDs.
- Python 3.14 Compatibility: pytorch-grad-cam incompatible. Solution: Rewrote using native torchvision transforms.
- Class Confusion: Passenger vs Safe has 19% confusion. Solution: Planned focal loss for hard examples.

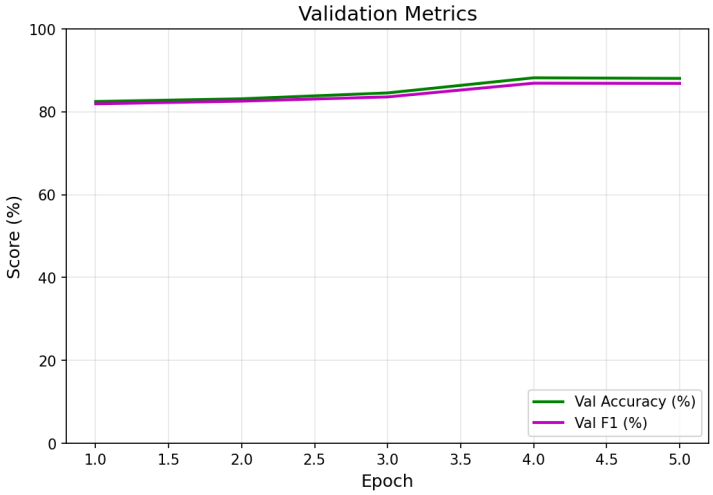
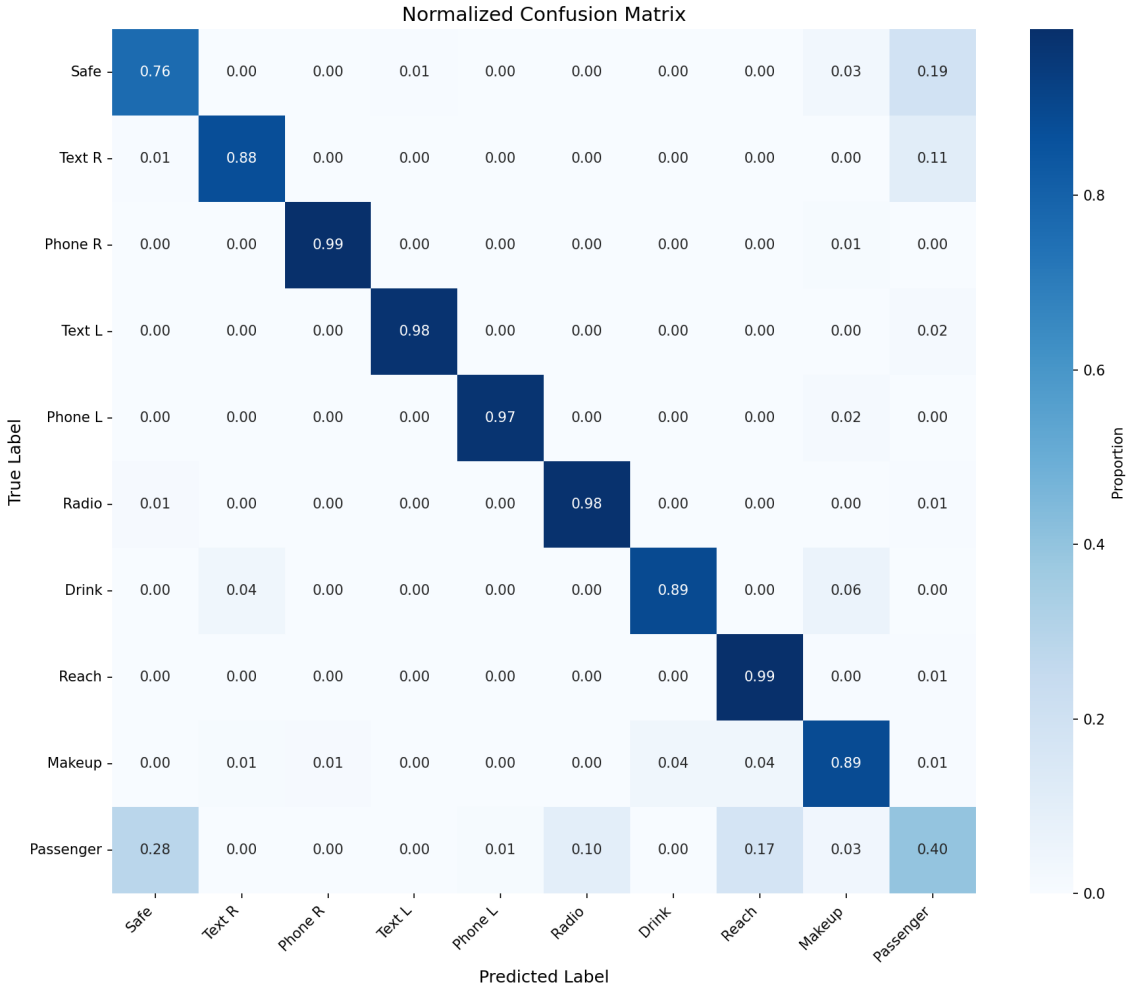
6. Next Steps for Final Phase

- Train ResNet18 and MobileNetV3 for architecture comparison
- Run zero-shot domain generalization on AUC dataset
- Complete CAM quality evaluation with Pointing Game metric
- Build Streamlit web demo for live inference
- Quantize ONNX model for faster edge deployment

Appendix: Training Results

Training Curves (Loss & Accuracy over 5 epochs):





Confusion Matrix (Normalized):